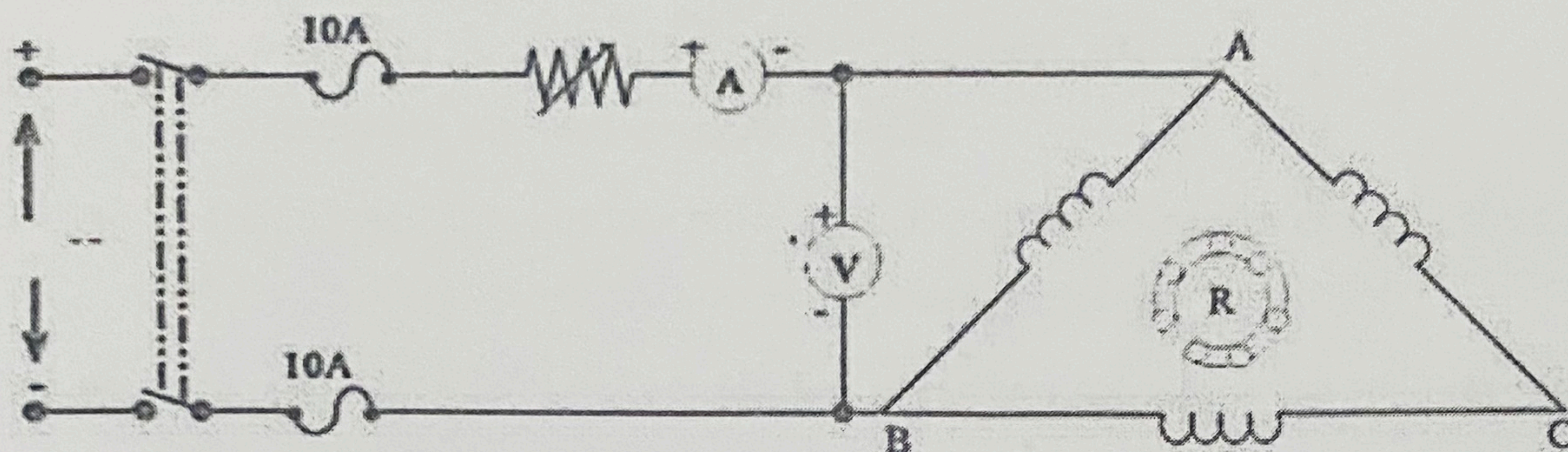


Switch on the 3 phase supply. Adjust the autotransformer so that rated current (to get full load copper loss) flows in the ammeter. Note down voltmeter, ammeter and wattmeter readings. (If any of the wattmeter reads negative, switch off the supply and interchange the connections of the pressure coil {or current coil} of that wattmeter and again take readings.)

C) STATOR RESISTANCE MEASUREMENT

Make the connections as shown in the diagram.



Precautions:

- Keep the rheostat in maximum resistance position

Switch on 30V d.c. supply. Note down voltmeter and ammeter readings for different positions of rheostat. (Note: Resistance/phase = $\frac{3}{2}$ Delta resistance.)

Procedure to draw the circle diagram : (Do not write in fair record)

13. Draw the lines by taking the current (I) in X-axis, voltage (V) in Y-axis. (V & I are line values).
14. From the No-load test find out the current I_0 and draw the OA vector with the magnitude of I_0 from the origin by suitable current scale, which lags the voltage (Y - axis) V by an angle ϕ_0 , where $\phi_0 = \cos^{-1} \left(\frac{W_{oc}}{\sqrt{3}V_{oc}I_{oc}} \right)$
15. From the current I_{sc} find out the I_{SN} (short circuit current corresponding to the normal voltage) through the formula $I_{SN} = I_{sc} \left(\frac{V_{rated}}{V_{sc}} \right)$, draw the OB vector with the magnitude of I_{SN} from the origin by the same current scale, which lags the voltage (Y-axis) V by an angle ϕ_{sc} where $\phi_{sc} = \cos^{-1} \left(\frac{W_{sc}}{\sqrt{3}V_{sc}I_{sc}} \right)$
16. Join the points B and A to get the output line.
17. Draw the parallel line for the X-axis from point A and for the Y-axis from point B upto the X-axis (point E), let both the lines intersect at point D.
18. Then draw the bisector for the output line and extend it to the line AD let the point of intersection be C.
19. By keeping the point C as center draw a semi circle with radius CA.